

INTERNSHIP REPORT

On

BEAST OS

A Cross-Platform EdTech Operating System by Beast Learners

Submitted To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.



By

Roll Number of Student: **CS-2341763**

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Section: **4CSE10**

Batch: 2023–27

CANDIDATE'S DECLARATION

I, Aditya Raj, do hereby declare that the internship report titled "Beast OS – A Cross-Platform EdTech Operating System for ICSE Board Preparation" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: Aditya Raj

Roll No.: CS-2341763

Date: 10/09/2026

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout my internship at Beast Learners. I am thankful for their aspiring guidance, invaluable constructive feedback, and friendly advice during the course of this work. I am sincerely grateful to my mentors and the engineering team at Beast Learners for sharing their honest and insightful views on the many front-end and UI engineering challenges I encountered while building the Beast OS platform.

I would like to thank my internship supervisor and the Beast Learners engineering team for the opportunity to work on the user-facing interface of a live, production-grade product, and for their continuous guidance on component design, responsive layout, and UI performance.

I would also like to express my sincere gratitude to the Department of Computer Science and Engineering, IILM University, Greater Noida, for facilitating this internship and for their academic support throughout the programme.

Finally, I express my heartfelt gratitude to my parents and friends for their constant encouragement and blessings.

Signature: _____

Student Name: Aditya Raj

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INTERNSHIP COMPLETION CERTIFICATE

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INTERNSHIP COMPLETION CERTIFICATE

September 10, 2026

This is to certify that **Aditya Raj** has successfully completed his three-month internship as a **Web Developer Intern** at **Beast Learners**, from **June 1, 2026 to September 1, 2026**.

During his internship, he contributed to the **development and enhancement of BeastOS**, the organization's digital platform, working on web development, feature implementation, and technical improvements. He demonstrated dedication, technical skills, and a strong willingness to learn throughout his internship.

We appreciate his valuable contributions to **Beast Learners** and wish him continued success in his future endeavors.

Sincerely,

Aditya Raj
Founder
Beast Learners

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4. PROJECT DESCRIPTION

4.1 Introduction

This report presents an account of the internship undertaken by Aditya Raj, a B.Tech Computer Science and Engineering student (Section 4CSE10) at IILM University, Greater Noida, U.P., at Beastlearners, an EdTech organisation building technology-driven solutions for school-level board examination preparation. The internship was carried out in the role of **Web Developer Intern**, with primary responsibility for engineering the student-facing web interface and the internal analytics dashboard of the organisation's flagship product.

The project undertaken during the internship, Beast OS, is a cross-platform EdTech Operating System designed to revolutionise ICSE Class 10 board examination preparation. The platform replaces static, print-based revision with an interactive web application in which students track their syllabus readiness, practise through AI-assisted tools, and generate subject-wise assessments, all from a single dashboard. Over the course of the internship, the intern was directly responsible for designing and building the front-end screens of BeastOS — the Home dashboard, the AI Tutor interface, the Quiz Generator, the Command Centre analytics view, and the Subscription Store — using React.js and a component-driven, responsive design system.

4.2 Organization Profile

Beastlearners is an EdTech organisation focused on reimagining how school students prepare for board-level examinations. The organisation's core product, Beast OS, positions itself as an "operating system" for board exam preparation rather than a conventional content-delivery app — combining structured syllabus tracking, AI tutoring, and gamified progress analytics into one platform. Beastlearners places strong emphasis on clean UI/UX, product velocity, and data-driven design, running an agile, feature-driven development process in which interns are given genuine ownership of production-facing screens rather than isolated training exercises.

The organisation's front-end engineering culture centres on a single-page application (BeastOS) built with React.js and Tailwind CSS, consumed by students on the web and mirrored administratively through a separate, permission-gated Admin Dashboard — giving the intern direct, hands-on exposure to real-world component architecture, state management, and dashboard/data-visualisation design.

4.3 Problem Statement

Traditional preparation for the ICSE Class 10 board examination remains largely static and fragmented. Students typically rely on printed material, generic mobile applications, and disconnected practice tests that provide no single, continuous, visual picture of syllabus readiness. Existing tools generally suffer from the following limitations:

- No single dashboard that consolidates a student's daily activity, streak, and syllabus progress in one glance.
- Quiz and practice tools that are not organised cleanly by subject and difficulty, making self-assessment tedious.
- No visual, at-a-glance breakdown of "board readiness" by subject and chapter, so students cannot easily identify their weakest topics.
- Subscription and monetisation screens that are cluttered or unclear about what each plan actually includes.
- No internal, real-time view for the organisation itself to track subscriber activity, credit usage, and plan distribution.

Beast OS was conceived to directly address these gaps by pairing a clean, card-based student dashboard with a dedicated analytics view ("Command Centre") for readiness tracking, and a parallel BI dashboard for internal monitoring — all built as a single, cohesive front-end system.

4.4 Project Objectives

- To design and build a unified Student Dashboard ("Home") consolidating daily challenges, study streaks, and quick-access navigation to core learning tools.
- To develop the front-end interface of the AI Tutor chat module, enabling a conversational, credit-based tutoring experience.
- To build a configurable Quiz Generator UI allowing students to select subject, chapter category, and quiz length before generating an assessment.
- To engineer the "Command Centre" — a data-visualisation dashboard displaying overall board readiness, per-subject health scores, and a prioritised list of the student's biggest topic-level gaps.
- To build the Subscription Store front end, presenting Monthly and Yearly Pro plans with clear feature comparisons and upgrade call-to-actions.
- To gain practical, production-level experience with component-based front-end architecture, responsive design, state-driven UI updates, and basic data-visualisation using chart libraries.

4.5 Scope of the Project

The scope of the internship project was focused on the front-end/UI layer of five modules within Beast OS – from wireframing through to responsive, production-ready implementation on the web client. The work covered React component design, Tailwind-based responsive layout, client-side state handling for dynamic widgets (streaks, progress bars, credit counters), and integration with existing backend APIs to render live data (readiness percentages, subscriber counts, chart data) rather than static mock-ups. The scope did not extend to backend/API development, database schema design, payment-gateway server-side logic, or infrastructure/DevOps ownership, which remained the responsibility of the backend and full-stack engineering staff.

4.6 Technologies and Tools Used

Table 4.1: Technology Stack Used in Beast OS (Web Developer Scope)

Layer	Technology / Tool	Purpose
Frontend (Web)	React.js, Tailwind CSS	Component-based, responsive UI for the student dashboard and admin panel
Data Visualisation	Chart.js / Recharts	Rendering the Daily Active Credits line chart and Subscription Plan Distribution bar chart
State & Data Fetching	React Hooks, REST API integration	Fetching and rendering live readiness scores, credits, and subscriber data
UI/Icons	Lucide icons, custom card components	Consistent iconography across Home, Quiz, and Command Centre screens
Backend (consumed, not built)	Supabase (PostgreSQL, Edge Functions)	Source of live data displayed on the dashboards
Payments (consumed, not built)	Razorpay API	Backing the Subscription Store's "Upgrade Now" actions

4.7 System Architecture

Beast OS follows a decoupled, client-server architecture in which the React.js web application acts as an independent presentation-layer client that communicates with a shared backend over REST APIs. As a Web Developer Intern, the focus was on this presentation layer: structuring reusable components (navigation sidebar, stat cards, progress bars, chart widgets) that consume data exposed by backend endpoints and render it consistently across the Student Dashboard, Command Centre, and Admin BI Dashboard.

The front end is organised as a single-page application with a persistent left-hand navigation shell ("BeastOS") common to Home, AI Tutor, Study Hub, Quiz, Subscription Store, and Command Centre. Business-critical calculations — such as readiness percentages, credit

balances, and subscription status — are computed on the server and simply rendered by the client, ensuring the UI always reflects a single, trusted source of truth rather than performing its own calculations.

Figure 4.1: High-Level Front-End Architecture of Beast OS

4.8 Methodology

The internship followed an agile, screen-by-screen methodology in which each interface was wireframed, built as reusable components, connected to live data, and shipped iteratively. The intern was responsible for engineering the following front-end modules of Beast OS, described below, followed by an account of key technical challenges and solutions implemented.

4.8.1 Student Home Dashboard

- Built the landing dashboard shown to students on login, including a personalised greeting, a "Daily Challenge" banner highlighting the student's weakest topic, and a live Study Streak counter.
- Designed a four-card quick-access grid (AI Tutor, Quiz, Notes, Library) using consistent icon-and-label components for fast navigation.
- Implemented the "Syllabus Progress" widget with an empty-state fallback ("No progress records found") for new users, and a populated state once course data exists.

4.8.2 AI Tutor Chat Interface

- Built the conversational AI Tutor screen, including the chat window, welcome message, and message input bar.
- Implemented the credit-balance indicator ("500 Credits") in the header so students can track their remaining AI usage at a glance.
- Ensured the chat UI gracefully handles empty and loading states before the first message is sent.

4.8.3 Quiz Generator

- Built the "Generate AI Quiz" screen allowing students to choose quiz length (5 / 10 / 15 / Custom questions) via a segmented control.
- Designed expandable subject cards (Physics, Chemistry, Biology, Mathematics, History & Civics, Geography, Computer Applications), each with its own icon and chapter-selection dropdown.
- Ensured the layout remained consistent and responsive across a two-column grid regardless of the number of subjects configured on the backend.

4.8.4 Command Centre – Board Readiness Analytics

- Built the "Board Readiness" hero banner displaying an overall percentage score and progress bar personalised to the logged-in student.
- Implemented the four-stat summary row (Mastered / In Progress / Needs Attention / Not Started) as reusable colour-coded stat cards.
- Built the "Subject Health" section with per-subject progress bars and chapter-readiness counts, and the "Biggest Gaps" panel listing the lowest-readiness topics with a "Fix Now" call-to-action button.

4.8.5 Subscription Store & Admin BI Dashboard

- Built the Subscription Store screen presenting Pro Monthly and Pro Yearly plans side by side, with pricing, feature checklists, and "Upgrade Now" buttons, including a highlighted savings badge on the annual plan.
- Contributed to the internal Admin "BI Analytics Dashboard," building the Active Subscribers, Credits Burned, and Live DAU stat cards, along with the Daily Active Credits line chart and Subscription Plan Distribution bar chart.

4.8.6 Challenges Faced and Solutions Implemented

Table 4.2: Key Technical Challenges and Solutions

Challenge	Solution Implemented
Rendering consistent card and chart components across two very different dashboards (Student Command Centre vs. Admin BI Dashboard) without duplicating code.	Built a shared library of reusable stat-card and progress-bar components parameterised by colour, label, and value, reused across both dashboards.
Handling empty/zero-data states gracefully (e.g., "0 Days" streak, "No progress records found," "0" DAU) without the UI looking broken for new users.	Designed explicit empty-state variants for every widget, matching the platform's visual language instead of leaving blank space.
Keeping the multi-subject Quiz Generator grid visually balanced when subjects have varying numbers of chapters/options.	Used a responsive two-column CSS grid with fixed-height cards and expandable dropdowns, so uneven content never breaks the layout.
Displaying live, frequently changing values (credits, DAU, readiness %) without the dashboard feeling stale or requiring a manual refresh.	Connected components to the backend's REST endpoints via React hooks so values update on each page load/re-fetch without hardcoding data into the UI.
Ensuring the "Biggest Gaps" and "Subject Health" panels stayed readable as the number of flagged topics grew.	Capped the visible list with a scroll/overflow pattern and colour-coded readiness percentages (red/amber) to prioritise attention at a glance.

4.9 Expected Outcomes

The internship resulted in five production-facing front-end modules being shipped as part of Beast OS, each directly improving the student and admin experience. The Home dashboard gave students a single, motivating entry point into daily study; the AI Tutor and Quiz Generator interfaces made core learning tools easy to discover and use; the Command Centre gave students

a clear, visual picture of their board-exam readiness; and the Subscription Store and Admin BI Dashboard supported the organisation's monetisation and internal monitoring needs.

On a personal level, the internship provided substantial hands-on experience with component-based front-end architecture using React.js, responsive design with Tailwind CSS, data-visualisation using chart libraries, and consuming live REST APIs to build dashboards that reflect real, changing data rather than static mock-ups. These outcomes align closely with industry expectations for a Web Developer role and have significantly strengthened the intern's readiness for professional front-end engineering work.

As future scope, the Command Centre could be extended with trend charts showing readiness improvement over time, the Quiz Generator could support drag-and-drop chapter reordering, and the Admin BI Dashboard could be extended with exportable, date-range-filtered analytics views.

4.10 Certificates of Completion and Communication Proof

The Internship Completion Certificate issued by Beastlearners is attached in the front matter of this report (see "Internship Completion Certificate"). Copies of communication and other proof of engagement with the organisation during the internship period are appended below.



5. BIBLIOGRAPHY / REFERENCES

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